A bioinformatics pipeline for the delineation of co-binding transcription factor conformations

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Introduction

Transcription factors (TFs) are key regulatory proteins for gene transcription that bind to short DNA sequences called “motifs” on non-coding regions of the genome. They largely depend on cooperative interactions with other TFs for the fine regulatory control they exhibit on gene expression, which also makes genetic variants in non-coding regions where TFs bind a prevalent and difficult to examine source of disease. A large part of this difficulty is the lack of well established, easily performed methods to characterize and better understand the interactions between co-binding TFs. Thus, we hypothesized that a systematic observation of the distances and orientation preferences between TF motifs in co-binding sites would be an effective way to characterize the interactions in any given pair while remaining simple to implement and perform.

Method

To this end, we designed a computational pipeline through which we could examine TF co-binding in the aforementioned manner. The pipeline works by taking experimental ChIP-seq narrowPeak files, intersecting them with provided co-binding TF peak files, then annotating each identified co-bound peak with a corresponding sequence on hg19. These sequences are then scanned for the motifs of the TFs present in those peaks, identifying the highest scoring one for each. The positions of these motifs are then used to calculate their distance and direction, and the strands the motifs are located on are then used to determine the orientation of the motifs in any given peak relative to each other. These motif distances are then plotted in a normalized histogram against randomized distances and then enriched to identify ranges with excessive binding. These are later stratified by orientation to identify whether preferences for a distance range correspond to preferences for one orientation over another. In order to test this pipeline, we examined six type 2 diabetes associated TFs, matched into 15 TF pairs.

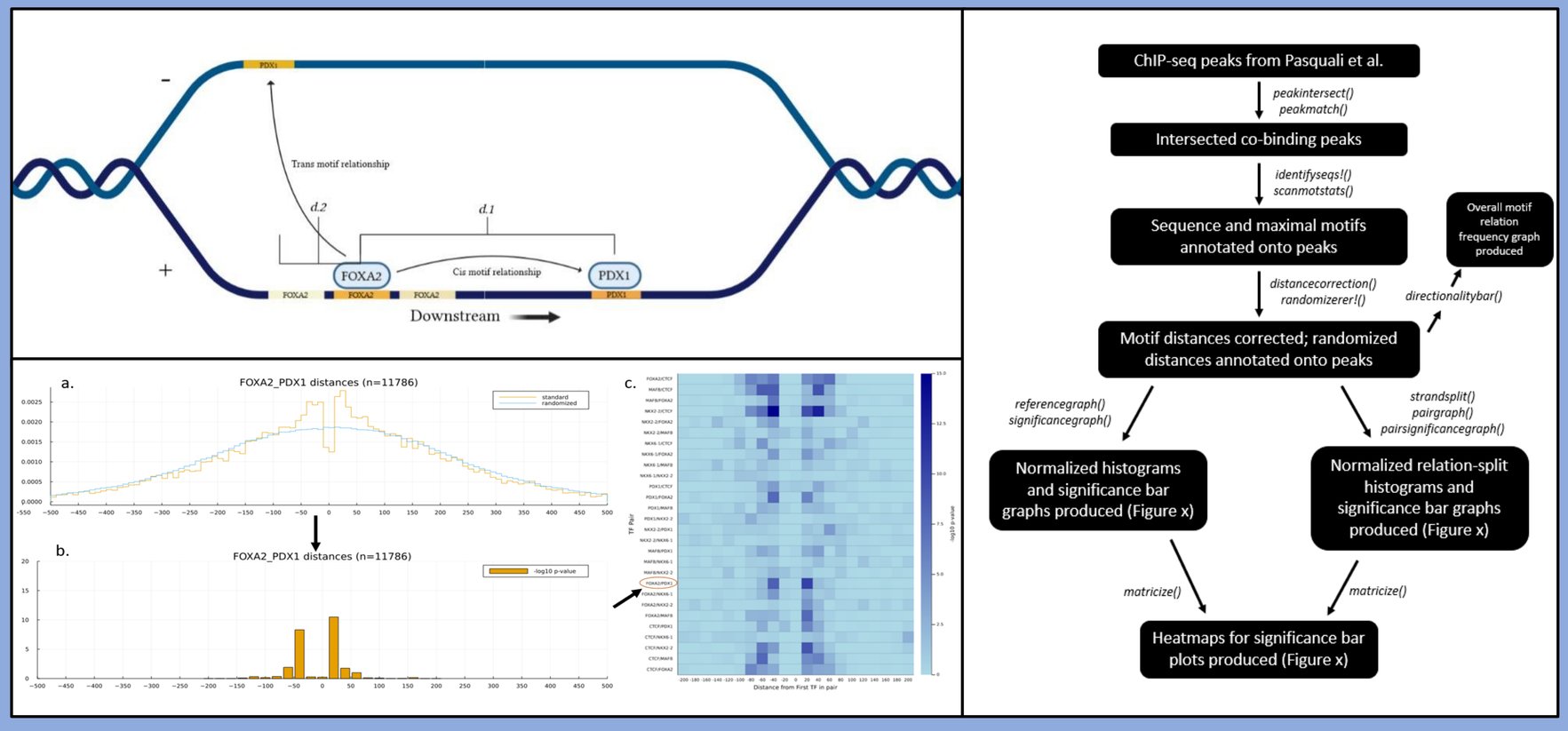

Figure 1: A visualization of the terminology and data flow used in this study.
A) In this visualization, we can see larger motif scores represented by darker shades, with respective TFs binding to the ones with the most similarity to its corresponding sequence. In this example, the d.1 distance would be a positive number to indicate a downstream direction from FOXA2 to PDX1, whereas d.2 would be negative, regardless of the strand it is located on. Lastly, “cis” and “trans” motif orientations are labelled as such, corresponding to co-bound motifs on the same and opposite strands respectively.
B) We can see the progression from normalized histograms to excess graphs to the heatmaps shown below. (a) Shows the normalized motif distance histograms we begin with, with the randomized reference distribution in blue and the observed distance distribution in yellow. (b) Shows the bar plots of -log10 p-value calculated for each bin calculated in reference to the randomized curve from the histogram above. (c) Shows the final step, in which each of the bar plots for every TF pair is plotted onto one heatmap for convenience and easier visualization.
C) The black boxes represent the current data at each stage, while the arrows represent their source and following output, annotated by the functions used to move from one set of data to another.

Results

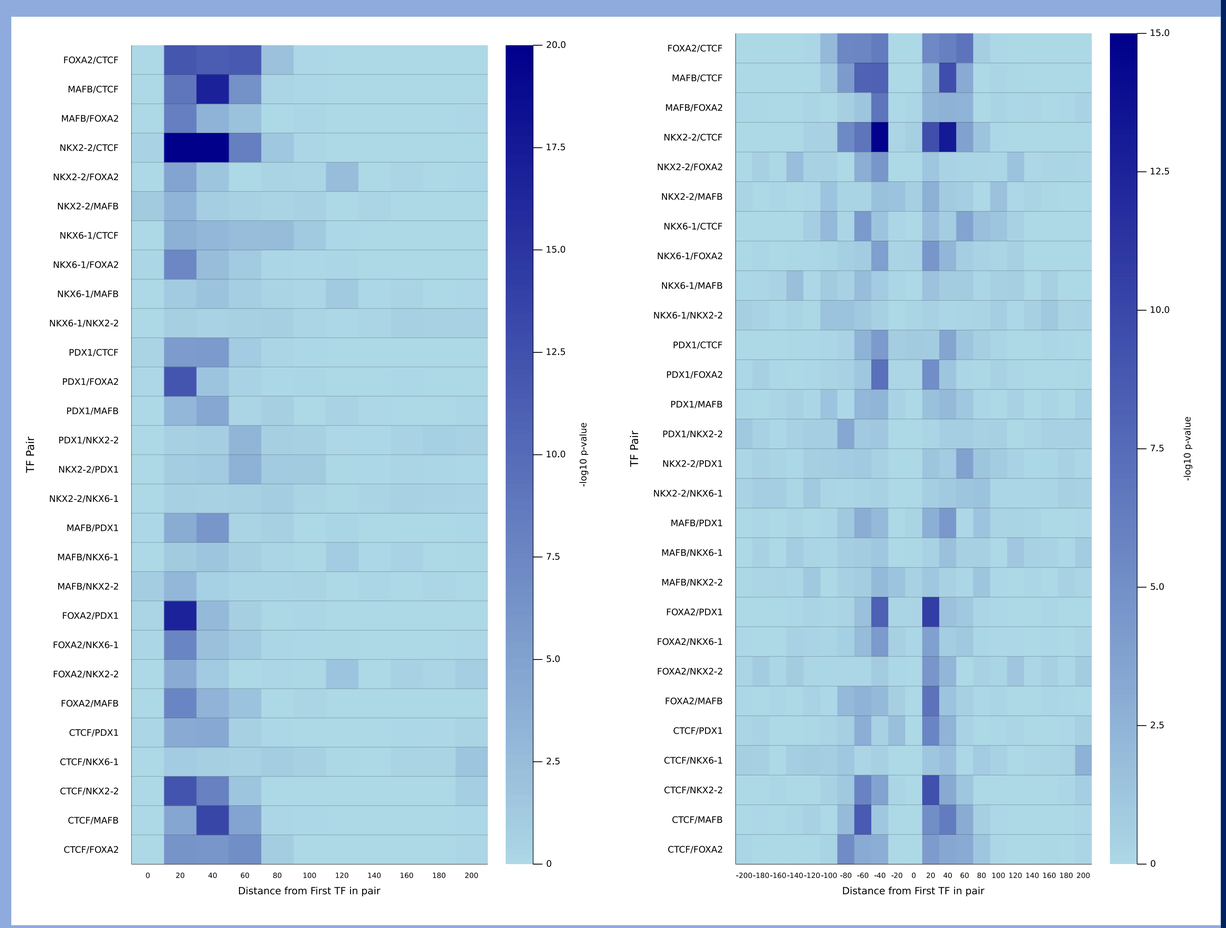

Figure 2: Absolute and bidirectional motif distance frequencies.
P-values for the frequencies with which specific distance ranges appear in each TF pair relative to what would be expected, multiplied by –log10. The -log10 p-value scale is capped at 20 and 15 respectively to avoid burying reasonably significant distance ranges under much stronger p-values. Notably, CTCF shows strong relationships with NKX2-2 and MAFB, and FOXA2 shows a strong relationship with PDX1, all three within specific ranges, with slight directional preferences.

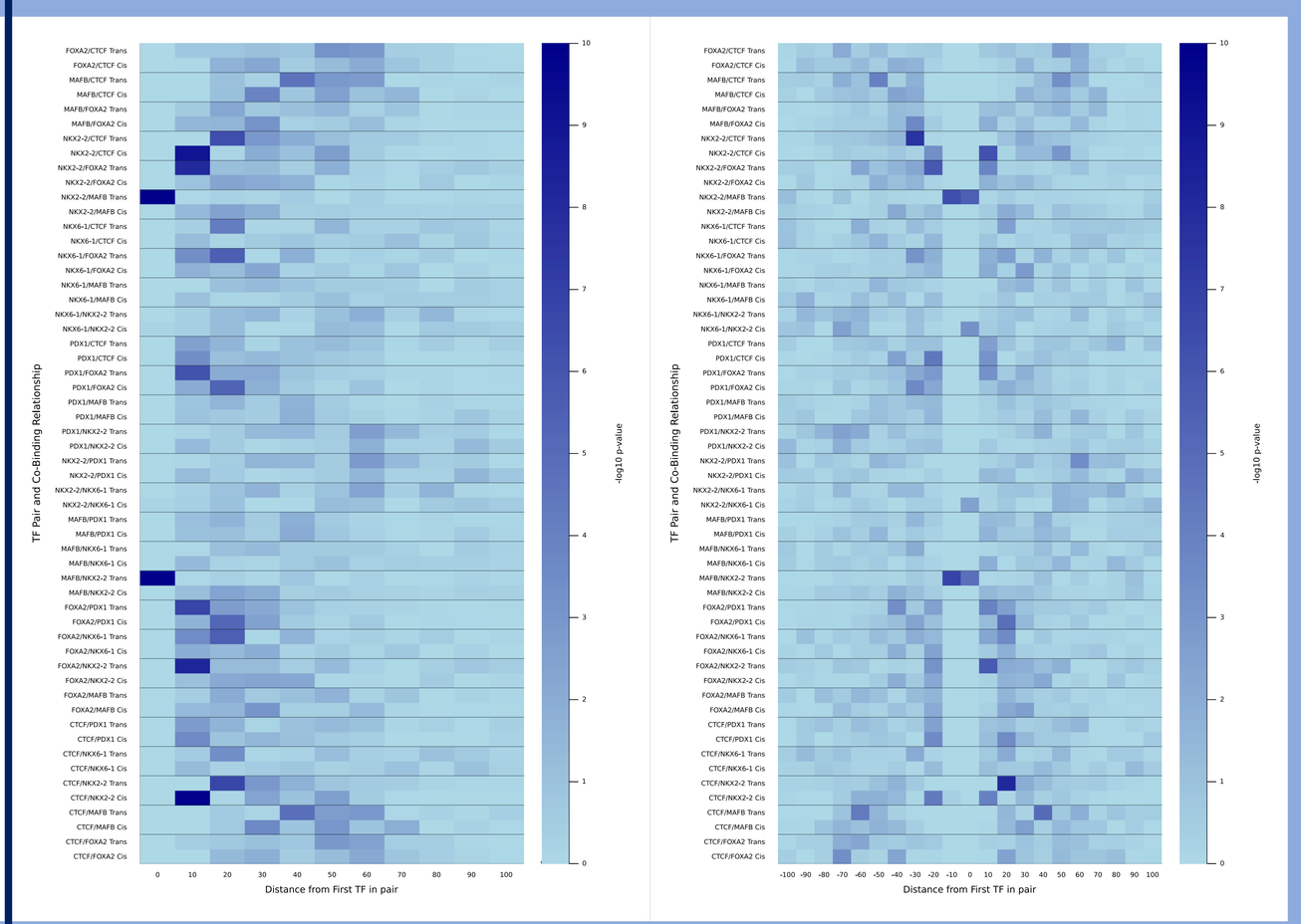

Figure 3: Absolute and bidirectional orientation-stratified motif distance frequencies .
P-values for the frequencies with which specific distance ranges appear in each TF pair relative to what would be expected, multiplied by –log10. The -log10 p-value scale is capped at 10. Notably, CTCF shows a bidirectional cis preference in the 10-20bp range but a strongly directional trans preference in the 20-30bp range. MAFB/NKX2-2 also show a significant preference for trans orientations in the 0-10bp range. FOXA2/NKX2-2 shows a strong trans preference in the 10-20bp range while FOXA2/PDX1 switches from a trans preference in the 10-20bp range to a cis preference at 20-30bp.

Conclusions

- NKX2-2 binding 20-60bp upstream from CTCF suggests their interaction depends on CTCF’s nucleosome arrangement influence
- MAFB/NKX2-2 0-9bp co-binding in trans orientations suggests a combined motif
- This pipeline can potentially help highlight areas of interest in cooperatively binding TFs in a quick and streamlined way to help identify areas for further research in any future TF systems